

Jinjin Zhao

✉ jinjinzhao24@gmail.com ☎ +1 312 358 4946 🔗 jinjinz.com 🌐 j2zhao

About

Research Statement: My research focuses on building systems for semi-structured machine learning data - spanning tools for large array operations, data science notebooks, and AI pipelines. Currently, I'm particularly interested in projects that coordinate data at scale (e.g., across distributed environments) or enable safe, automatic feedback loops between data generation and use (e.g., agentic pipelines).

Key Words: AI (LLM) Orchestration, Distributed Systems, Machine Learning Infrastructure, Data Platforms, Databases

Languages and Skills: Python, C, PyTorch, Airflow, Amazon Web Services, Google Cloud Platform, Kubernetes, SQL

Education

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| PhD | University of Chicago , Computer Science | Sept. 2019 to Dec. 2025 (est.) |
| | <ul style="list-style-type: none">• Advisor: Sanjay Krishnan (ChiData Database Group)• M.S. degree completed as part of Ph.D. program | |
| BSE | Princeton University , Computer Science | Sept. 2015 to May 2019 |
| | <ul style="list-style-type: none">• Summa Cum Laude• Minor in Statistics and Machine Learning | |

Selected Projects

TableVault: Contextual End-to-End Management of LLM Data Pipelines ([Github](#)) (On-going)

- A open-sourced failure-safe data store for distributed AI notebooks and Python processes backed by ArangoDB.
- Enables coordinator-less control where workers can communicate execution requests (e.g. pause/resume/stop) via shared state.
- Supports rich contextual search where users can issue joint queries over AI artifacts (e.g. documents, embeddings, dataframes), and pipeline context (e.g. code, prompts, comments).

DSLog: A Compressed Query and Storage Framework for Fine-Grained Array Lineage

- Novel framework for automatically converting cell-level lineage to queryable geometric representations.
- Augmented the Numpy library with low-level memory management (C) to record cell-to-cell operational lineage efficiently, resulting in up to 3000x improvements over Python baselines.
- Designed a new range-based compression algorithm that improves storage space and query time of the resulting lineage graph by up to 2000x and 20x respectively.

Tracing Variation in Data Science Workflows with Jupyter Notebook Logging

- First study to quantitatively assess common conceptions in data science (e.g. data cleaning is 80% of the work).
- Designed a tool to log Jupyter notebook execution traces from 97 novices and industry experts in data science.
- Compared string search, code analysis (AST) and notebook replay techniques to analyse JSON traces for capturing user trends (e.g. most errors are resolved within 1-2 code executions).

Experience

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|--|-------------------------|
| Linea Labs , Research Intern | CA, USA |
| <ul style="list-style-type: none">• Co-designed initial MVP for Airflow reproducibility with a 6-person startup team.• Implemented novel lineage-tracking and LLM-based code-analysis features. | June 2023 to Sept. 2023 |
| Princeton Plasma Physics Lab , Research Intern | NJ, USA |
| <ul style="list-style-type: none">• Compiled data on two years of DIII-D tokamak experiments and trained neural network models to predict pedestal features driving fusion output. | June 2018 to July 2018 |

Meta, Software Engineer Intern

- Developed a full-stack Hack (PHP) and MySQL tool for storing and retrieving test artifacts (e.g., logs and build files) from internal pre-commit automated runs.
- Scaled to over 200 million files per week, supporting the majority of internal code development.

WA, USA
June 2017 to Aug. 2017

Meta, Facebook University Intern

- Designed and built an independent Android app with an internal music player that generated Spotify playlists based on nearby concerts.

CA, USA
June 2016 to Aug. 2016

Publications

Fast Capture of Cell-Level Provenance in Numpy.

2025

Jinjin Zhao, Sanjay Krishnan

ProvenanceWeek@SIGMOD [Paper](#) 

TableVault: Managing Dynamic Data Collections for LLM-Augmented Workflows

2025

Jinjin Zhao, Sanjay Krishnan

NOVAS@SIGMOD [Paper](#) 

Quantifying Variation in Data Science with Fine-Grained Procedural Logging.

2024

Jinjin Zhao, Avigor Gal, Sanjay Krishnan

Under Submission [Paper](#) 

Compression and In-Situ Query Processing for Fine-Grained Array Lineage.

2024

Jinjin Zhao, Sanjay Krishnan

ICDE [Paper](#) 

Data Makes Better Data Scientists.

2023

Jinjin Zhao, Avidgor Gal, Sanjay Krishnan

HILDA@SIGMOD [Paper](#) 

AMIR: Active Multimodal Interaction Recognition from Video and Network Traffic in Connected Environments.

2023

Shinan Liu, Tarun Mangla, Ted Shaowang, **Jinjin Zhao**, Sanjay Krishnan, Nick Feamster

UbiComp/IMWUT [Paper](#) 

Data Station: Delegated, Trustworthy, and Auditable Computation to Enable Data Sharing Consortia with a Data Escrow.

2022

Siyuan Xia, Zhiru Zhu, Chris Zhu, **Jinjin Zhao**, Kyle Chard, Aaron J. Elmore, Ian Foster, Michael Franklin, Sanjay Krishnan, Raul Castro Fernandez

VLDB [Paper](#) 

Towards Causal Query Answering for Debugging Video Analytics Systems.

2022

Ted Shaowang*, **Jinjin Zhao***, Stavros Sintos, Sanjay Krishnan

HILDA@SIGMOD [Paper](#) 

Prediction of DIII-D Pedestal Structure From Externally Controllable Parameters.

2021

Emi Zeger, Florian Laggner, Alessandro Bortolon, Cristina Rea, Orso Meneghini, Samuli Saarelma, Brian Sammuli, Sterling Smith, **Jinjin Zhao**

IEEE Transactions on Plasma Science [Paper](#) 

Activities And Awards

- **2018 - 2024 Teaching Assistant:** COS 397/497 Fall 2018 (*Princeton University*), CMSC 16100 Autumn 2019 (*University of Chicago*), CMSC 21800 Autumn 2020/2023 (*University of Chicago*), DATA 13600 Spring 2024 (*University of Chicago*)
- 2020 - 2022 Curriculum and Social Minister, UChicago CS Graduate Student Ministry
- ICDE'2024 Travel Award, NSF
- 2022 University Unrestricted Fellowship, University of Chicago
- 2020 Lab Coordinator, CDAC (high school data science research summer program)
- OSDI'2020 Diversity Grant, USENIX Association
- 2019 Neubauer Graduate Scholarship, University of Chicago